On Runs of Integers with Large Prime Factors: A Lower Bound for $k(n)$ and the Status of the Upper Bound

April 2026

Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$k(n) \;\geq\; \exp\bigl((\log n)^{1/2-\varepsilon}\bigr)$$

for every fixed $\varepsilon > 0$ and all sufficiently large n, using Rankin's method and a pigeonhole argument.

We also discuss the upper bound $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$, which would give $\log k(n) = (\log n)^{1/2+o(1)}$. The upper bound argument requires the existence of a k_1-smooth integer in every interval $[m+1, m+k_1]$ with $m \leq n$ and $k_1 = \exp((\log n)^{1/2+\varepsilon})$. We reduce this to a short-interval smooth number estimate (Hypothesis 5 below) and prove the upper bound *conditional on that hypothesis*. We discuss what the available literature (Hildebrand–Tenenbaum [1], Tenenbaum [2]) actually establishes and where the gap lies.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Set

$$\Psi(x, y) := \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. A *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \le n$ and $P^+(m+j) > k$ for all $1 \le j \le k$. Define $k(n)$ to be the maximum k for which a k-run below n exists.

2 The Rankin Bound

Lemma 3 (Rankin bound [2, Theorem III.5.1])**.** *There is an effectively computable absolute constant $A \ge 3$ such that for all $x \ge y \ge 2$ with $v := \log x/\log y \ge A$,*

$$\Psi(x,y) \;\le\; x \cdot e^{-\frac{1}{2} v \log v}.$$

Proof. Cited from [2, Theorem III.5.1]. Rankin's method with the choice $\sigma = 1 - (\log v)/\log y$ yields $\Psi(x,y) \le x\, e^{-v\log v + C_0 v/\log v}$ for an absolute constant C_0. The correction term $C_0 v/\log v \le \frac{1}{2} v \log v$ once $\log^2 v \ge 2C_0$, which holds for all $v \ge A$ with $A = e^{\sqrt{2C_0}}$. The full proof with all constants is in [2, Theorem III.5.1]. □

3 The Lower Bound

Theorem 4 (Lower bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \ge \exp\big((\log n)^{1/2-\varepsilon}\big)$.*

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: k_0-smooth numbers are sparse in $[1, n]$. $\mathrm{v}_0 := \log n/\log k_0 . Since \log k_0 = (\log n)^{1/2-\varepsilon}(1+o(1))$ we get $v_0 = (\log n)^{1/2+\varepsilon}(1+o(1)) \to \infty$. In particular $v_0 \ge A$ (the constant from Lemma 3) for all large n, so

$$\Psi(n, k_0) \;\le\; n \cdot e^{-\frac{1}{2} v_0 \log v_0}.$$

We estimate the exponent:

$$\tfrac{1}{2} v_0 \log v_0 = \tfrac{1}{2}(\log n)^{1/2+\varepsilon}(1+o(1)) \cdot (1/2+\varepsilon) \log\log n\,(1+o(1)) \;\ge\; (\log n)^{1/2+\varepsilon/2}$$

for all large n. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$, this gives

$$e^{-\frac{1}{2} v_0 \log v_0} \le e^{-(\log n)^{1/2+\varepsilon/2}} \ll n^{-1} \cdot k_0^{-1},$$

so in particular $\Psi(n, k_0) \le n/(2k_0)$ for all sufficiently large n.

Step 2: Pigeonhole gives an empty block. Let $B := \lfloor n/k_0 \rfloor$. Partition $\{1, \ldots, Bk_0\}$ into B blocks $I_i := [(i-1)k_0 + 1, ik_0]$. Every k_0-smooth integer in $\{1, \ldots, n\}$ lies in one of these blocks, and

$$\Psi(n, k_0) \ \leq \ \frac{n}{2k_0} \ \leq \ \frac{B+1}{2} \ < \ B$$

for all $n \geq 2k_0$ (i.e. all large n). By pigeonhole, some block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. For that i, every integer in I_i has largest prime factor $> k_0$. Setting $m = (i-1)k_0 \leq n$, the interval $\{m+1, \ldots, m+k_0\}$ is a k_0-run below n, so $k(n) \geq k_0 = \exp((\log n)^{1/2-\varepsilon}(1 + o(1)))$. □

4 The Upper Bound: Reduction and Gap

What needs to be proved

Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. The upper bound $k(n) \leq k_1$ is equivalent to: every interval $[m+1, m+k_1]$ with $m \leq n$ contains at least one k_1-smooth integer.

For $m < k_1$ this is immediate: $k_1 \in [m+1, m+k_1]$ and k_1 is k_1-smooth.

For $m \in [k_1, n]$, we need the following estimate.

Hypothesis 5 (Short-interval smooth numbers)**.** Let ρ denote the Dickman function and let $\varepsilon > 0$ be fixed. There exists $n_* = n_*(\varepsilon)$ such that for all $n \geq n_*$, all $m \in [k_1, n]$ with $k_1 = \exp((\log n)^{1/2+\varepsilon})$,

$$\Psi(m + k_1,\, k_1) \ - \ \Psi(m,\, k_1) \ \geq \ 1. \tag{1}$$

Theorem 6 (Upper bound, conditional)**.** *Assume Hypothesis 5. For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$.*

Proof. With Hypothesis 5 every interval $[m+1, m+k_1]$ with $m \in [k_1, n]$ contains a k_1-smooth integer, and the case $m < k_1$ was handled above. Hence no k_1-run below n exists, so $k(n) < k_1$. □

Discussion: what the literature provides

The main result of Hildebrand–Tenenbaum [1] is a uniform asymptotic for the *global count* $\Psi(x, y)$. Specifically, fixing $\varepsilon > 0$ and writing $\mathcal{R}_\varepsilon = \{(x, y) : x \geq y \geq 2,\ \log y \geq (\log\log x)^{5/3+\varepsilon}\}$, the result [1, Théorème 1] states

$$\Psi(x, y) = x\rho(u)\big(1 + \eta(x, y)\big), \tag{2}$$

where $\eta(x, y) \to 0$ uniformly for $(x, y) \in \mathcal{R}_\varepsilon$ as $y \to \infty$, where $u = \log x/\log y$.

One can formally subtract (2) at $x + y$ and at x to obtain

$$\Psi(x + y, y) - \Psi(x, y) = (x + y)\rho(u')\big(1 + \eta_1\big) - x\rho(u_1)\big(1 + \eta_2\big), \tag{3}$$

where $u' = \log(x + y)/\log y$ and $u_1 = \log x/\log y = u$. However, bounding (3) from below is *not* immediate because $u' - u_1 = \log(1 + y/x)/\log y$, and ρ has a derivative that needs to be controlled.

The standard estimate one needs is:

$$(x + y)\rho(u') - x\rho(u_1) = y\rho(u_1)\big(1 + O(1/\log y)\big), \tag{4}$$

which holds when $x \geq y$ (so $y/x \leq 1$) and $y \to \infty$. In our application $x = m \geq k_1 = y$, so this condition is met.

Remark 7. More precisely, the derivation of (4) from (2) uses the mean-value form

$$\rho(u') = \rho(u_1) + \rho'(\xi)(u' - u_1), \quad \xi \in [u_1, u'],$$

with $\rho'(\xi) = -\rho(\xi - 1)/\xi$ and $u' - u_1 = O(y/x \cdot 1/\log y)$. This gives

$$(x+y)\rho(u') - x\rho(u_1) = y\rho(u_1) + (x+y)\rho'(\xi)(u' - u_1) = y\rho(u_1)\Big(1 + O\Big(\frac{\rho(\xi - 1)}{\rho(u_1)} \cdot \frac{x + y}{y} \cdot \frac{y}{x\log y}\Big)\Big).$$

Since $\xi \geq u_1$ and ρ is decreasing, $\rho(\xi - 1)/\rho(u_1) \leq \rho(u_1 - 1)/\rho(u_1)$, which is bounded by $1/\rho_0$ for a computable constant (the ratio $\rho(u - 1)/\rho(u)$ is bounded in terms of u; see [2, Corollary III.5.4]). The factor $(x + y)/x = 1 + y/x \leq 2$ (since $x \geq y$). Hence the error is $O(1/\log y) \to 0$. This calculation is elementary given (2), but it does not appear to be written out explicitly in [1] or [2] in the exact form needed here. Until a published source states (1) directly in this parameter range, Hypothesis 5 remains unverified.

Remark 8. Assuming the derivation in Remark 7 is accepted, one can verify that Hypothesis 5 holds for our parameters as follows. Set $u_1 = \log m / \log k_1 \in [1, (\log n)^{1/2-\varepsilon}]$. From (4) and (2),

$$\Psi(m + k_1, k_1) - \Psi(m, k_1) = k_1 \rho(u_1)\big(1 + \eta\big), \quad |\eta| \le \tfrac{1}{2},$$

for all large n (the $\frac{1}{2}$ coming from both $\eta_1, \eta_2 \to 0$ and the error in (4)). The main term satisfies $k_1 \rho(u_1) \ge 2$ for all large n (as computed in detail in the proof of Theorem 4 with k_1 in place of k_0, which gives a positive lower bound on $\log(k_1 \rho(u_1))$). Hence the count is ≥ 1.

5 Main Result

Theorem 9. (a) *(Unconditional) For every fixed $\varepsilon > 0$ and all large n: $k(n) \ge \exp((\log n)^{1/2-\varepsilon})$.*

(b) *(Conditional on Hypothesis 5) For every fixed $\varepsilon > 0$ and all large n: $k(n) \le \exp((\log n)^{1/2+\varepsilon})$.*

Hence, conditional on Hypothesis 5, $\log k(n) = (\log n)^{1/2+o(1)}$. Hypothesis 5 is a short-interval smooth number estimate plausibly within reach of the Hildebrand–Tenenbaum method but not verified directly from the published sources [1, 2] here.

Proof. Part (a) is Theorem 4. Part (b) is Theorem 6. □

References

[1] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.

[2] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.